
1° Deck
Damage Report
2° Deck
NEW YORK
RISK
Arrival
DELAYED
18
Dexa melhorar um pouco
Nunca Lembra

5.1 - HTN SOA

1) Leaky steam pipe:
2017 ACC/AHA definitions for HTN:

- Normal: SBP < 120, DBP < 80
- Elevated BP: SBP 120-129, DBP < 80
- Stage 1 HTN: SBP 130-139 or DBP 80-89
- Stage 2 HTN: SBP >140 or DBP >90

2) Normal pressure gauge:
2017 ACC/AHA definitions for HTN:

- **Normal: SBP < 120, DBP < 80**
- Elevated BP: SBP 120-129, DBP < 80
- Stage 1 HTN: SBP 130-139 or DBP 80-89
- Stage 2 HTN: SBP >140 or DBP >90

3) Elevated pressure gauge:
2017 ACC/AHA definitions for HTN:

- Normal: SBP < 120, DBP < 80
- **Elevated BP: SBP 120-129, DBP < 80**
- Stage 1 HTN: SBP 130-139 or DBP 80-89
- Stage 2 HTN: SBP >140 or DBP >90

4) Stage 1 pressure gauge:
2017 ACC/AHA definitions for HTN:

- Normal: SBP < 120, DBP < 80
- Elevated BP: SBP 120-129, DBP < 80
- **Stage 1 HTN: SBP 130-139 or DBP 80-89**
- Stage 2 HTN: SBP >140 or DBP >90

5) Stage 2 pressure gauge:
2017 ACC/AHA definitions for HTN:

- Normal: SBP < 120, DBP < 80
- Elevated BP: SBP 120-129, DBP < 80
- Stage 1 HTN: SBP 130-139 or DBP 80-89
- **Stage 2 HTN: SBP >140 or DBP >90**

6) Contemplative young doctor:
White Coat Hypertension = high blood pressure isolated to the doctor's office

7) Dreaming of home:
Diagnose white coat hypertension by looking for **normal BP measurements at home**

8) Screen partition:
HTN screening is recommended for ALL patients 18+ every year, OR every 6 months for patients w/ risk factors or SBP 120-129

9) Secret 18th birthday party:
HTN screening is recommended for **ALL patients 18+ every year**, OR every 6 months for patients w/ risk factors or SBP 120-129

10) Guy with cake slice leaning on RISKy pipe:
HTN screening is recommended for ALL patients 18+ every year, OR every 6 months for patients w/ SBP 120-129 or HTN risk factors

11) Blood pressure cuff sleeves:
BP should be measured with patient comfortably seated, arm at heart level, & appropriately sized cuff → **measure BP in both arms initially**

12) Primary deck:
Most cases of HTN are **Primary HTN**, formerly called Essential HTN

13) Secondary deck:
Secondary HTN = hypertension due to an underlying medical condition

14) Red renal artery rope:
Causes of 2° HTN include **renal artery stenosis**, renal parenchymal disease, primary aldosteronism, Cushing's syndrome, pheochromocytoma, hypo/hyperthyroidism, hyperparathyroidism, & coarctation of the aorta

15) Kidney-shaped pulley:
Causes of 2° HTN include renal artery stenosis, **renal parenchymal disease**, primary aldosteronism, Cushing's syndrome, pheochromocytoma, hypo/hyperthyroidism, hyperparathyroidism, & coarctation of the aorta

16) MINERAL crystal chandelier:
Causes of 2° HTN include renal artery stenosis, renal parenchymal disease, **primary aldosteronism**, Cushing's syndrome, pheochromocytoma, hypo/hyperthyroidism, hyperparathyroidism, & coarctation of the aorta

17) Red cushion:
Causes of 2° HTN include renal artery stenosis, renal parenchymal disease, primary aldosteronism, **Cushing's syndrome**, pheochromocytoma, hypo/hyperthyroidism, hyperparathyroidism, & coarctation of the aorta

18) Rainbow renal shave ice with adrenal toppings:
Causes of 2° HTN include renal artery stenosis, renal parenchymal disease, primary aldosteronism, Cushing's syndrome, **pheochromocytoma**, hypo/hyperthyroidism, hyperparathyroidism, & coarctation of the aorta

19) Man with undone bowtie:
Causes of 2° HTN include renal artery stenosis, renal parenchymal disease, primary aldosteronism, Cushing's syndrome, pheochromocytoma, **hypo/hyperthyroidism**, hyperparathyroidism, & coarctation of the aorta

20) Parathyroid buttons:
Causes of 2° HTN include renal artery stenosis, renal parenchymal disease, primary aldosteronism, Cushing's syndrome, pheochromocytoma, hypo/hyperthyroidism, **hyperparathyroidism**, & coarctation of the aorta

21) Pipe with constriction:
Causes of 2° HTN include renal artery stenosis, renal parenchymal disease, primary aldosteronism, Cushing's syndrome, pheochromocytoma, hypo/hyperthyroidism, hyperparathyroidism, & **coarctation of the aorta**

22(S) Happy man:
Patients w/ HTN are often **asymptomatic (Chief Complaint)**

23(S) Seasoned veteran of the seas:
HTN risk factors include **older age, male sex, African-American race, obesity**, OSA, & family history of HTN **(HPI)**

4(S) Captain asleep at the wheel:
HTN risk factors include older age, male sex, race, obesity, **OSA**, & family history of HTN **(PMHx)**

25(S) Family portrait:
HTN risk factors include older age, male sex, race, obesity, OSA, & family history of HTN **(Family Hx)**

26(S) Salty peanut snacks:
Lifestyle risk factors for HTN include a **high sodium diet** & alcohol use **(Social Hx)**

27(S) Bottle of booze:
Lifestyle risk factors for HTN include a high sodium diet & **alcohol use (Social Hx)**

28(S) Damage report:
Ask about markers of **end-organ damage** (i.e., CAD, ACS, heart failure, stroke, CKD) **(PMHx)**

29(S) Home med bag:
Ask about **home medications** that can contribute to HTN (e.g., OCPs, nasal decongestants, systemic glucocorticoids, NSAIDs) **(Medications)**

30(S) OCP ship's wheel:
Ask about home medications that can contribute to HTN (e.g., **OCPs**, nasal decongestants, systemic glucocorticoids, NSAIDs) **(Medications)**

31(S) Nasal spray bottle:
Ask about home medications that can contribute to HTN (e.g., OCPs, **nasal decongestants**, systemic glucocorticoids, NSAIDs) **(Medications)**

32(S) Moon-face porthole:
Ask about home medications that can contribute to HTN (e.g., OCPs, nasal decongestants, **systemic glucocorticoids**, NSAIDs) **(Medications)**

33(S) Fire extinguisher:
Ask about home medications that can contribute to HTN (e.g., OCPs, nasal decongestants, systemic glucocorticoids, **NSAIDs) (Medications)**

34(S) Cup of hot cocoa:
Stimulant drugs like **cocaine** and methamphetamine can also cause HTN **(Social Hx)**

35(S) Young boy running:
Consider secondary HTN in **young patients**, older patients with new-onset hypertension, or patients w/ formerly well-controlled BP that acutely worsens **(HPI)**

36(S) Old man with steam out of ears:
Consider secondary HTN in young patients, **older patients with new-onset hypertension**, or patients w/ formerly well-controlled BP that acutely worsens **(HPI)**

37(S) Ringing alarm clock:
Consider secondary HTN in young patients, older patients with new-onset hypertension, or **patients w/ formerly well-controlled BP that acutely worsens (HPI)**

38(S) Older man pulling renal artery rope:
RAS in older patients is most commonly caused by atherosclerosis → ask about hx of CAD, PAD, or stroke **(PMHx)**

39(S) Young, muscular woman pulling renal artery rope:
RAS in young women <35 is most commonly caused by **fibromuscular dysplasia (HPI)**

40(S) Brain hat on young muscular woman:
Fibromuscular dysplasia can also cause **neurological symptoms** 2/2 involvement of carotid & vertebral arteries **(ROS)**

41(S) Wet life vests on rope pullers:
RAS can present with recurrent episodes of flash **pulmonary edema (ROS)**

42(S) Diner with brain freeze & sweating:
Pheochromocytoma often presents with episodes of **diaphoresis and headache (ROS)**

43(S) Woman on luxurious cushion:
Patients with HTN 2/2 **Cushing's syndrome** will also present with weight gain, menstrual irregularities, and hirsutism **(ROS)**

44(O) Retina searchlight:
Hypertensive retinopathy flame hemorrhages, produces arteriovenous nicking, & cotton-wool spots on fundoscopy **(Physical Exam)**

45(O) Flaming searchlight with crossed wires:
Hypertensive retinopathy produces **flame hemorrhages, arteriovenous nicking, & cotton-wool spots** on fundoscopy **(Physical Exam)**

46(O) 4-shaped chair:
Listen for an **S4 gallop** on cardiac auscultation, which can suggest left ventricular hypertrophy **(Physical Exam)**

47(O) LARGE left epaulette:
Listen for an S4 gallop on cardiac auscultation, which can suggest **left ventricular hypertrophy** **(Physical Exam)**

48(O) Pinched-off pipe:
Coarctation of the aorta diagnosed in adulthood usually presents w/ HTN proximal to the coarctation, normal-low BP distal to the coarct, & systolic murmur **(Physical Exam)**

49(O) Murmur lines:
Coarctation of the aorta diagnosed in adulthood usually presents w/ **systolic murmur**, HTN proximal to the coarctation, & low BP distal to the coarct **(Physical Exam)**

50(O) High pressure gauge:
Coarctation of the aorta diagnosed in adulthood usually presents w/ systolic murmur, **HTN proximal to the coarctation**, & low BP distal to the coarct **(Physical Exam)**

51(O) Low pressure gauge:
Coarctation of the aorta diagnosed in adulthood usually presents w/ systolic murmur, HTN proximal to the coarctation, & **low BP distal to the coarct (Physical Exam)**

52(O) Arrival DELAYED:
Brachial-femoral pulse delay is a classic finding in significant coarctation of the aorta **(Physical Exam)**

53(O) Noisy red pipes:
Listen for an **abdominal bruit** to clue you into renal artery stenosis **(Physical Exam)**

54(O) Bunches of kidney balloons:
Bilateral upper abdominal masses in a patient with HTN suggests **autosomal dominant polycystic kidney disease** **(Physical Exam)**

55(O) Obese lady sitting on cushion:
Cushing's syndrome findings include **truncal obesity**, moon facies, proximal muscle weakness, & abdominal striae **(Physical Exam)**

56(O) Obese lady's round face:
Cushing's syndrome findings include truncal obesity, **moon facies**, proximal muscle weakness, & abdominal striae **(Physical Exam)**

57(O) Obese lady's thin arms:
Cushing's syndrome findings include truncal obesity, moon facies, **proximal muscle weakness**, & abdominal striae **(Physical Exam)**

58(O) Reddish-purple striated curtains:
Cushing's syndrome findings include truncal obesity, moon facies, proximal muscle weakness, & **abdominal striae (Physical Exam)**

59(O) Blue sweet candy:
Order **HbA1c** & fasting lipid panel to assess for comorbid type 2 diabetes and hyperlipidemia **(Labs)**

60(O) Fatty, cheesy pasta plate:
Order HbA1c & **fasting lipid panel** to assess for comorbid type 2 diabetes and hyperlipidemia **(Labs)**

61(O) Fishbone-esque dinner tray:
A basic metabolic panel, calcium level, & TSH are recommended for all hypertensive patients **(Labs)**

62(O) Calci-yum ice cream:
A basic metabolic panel, **calcium level**, & TSH are recommended for all hypertensive patients **(Labs)**

63(O) Waiter's red bow tie:
A basic metabolic panel, calcium level, & **TSH** are recommended for all hypertensive patients **(Labs)**

64(O) Pitcher of lemonade:
Order a **urinalysis** to assess for kidney disease, which can be a cause OR effect of HTN **(Labs)**
Symbols

65(O) ECG tablecloth:
Order an **ECG** to assess for LV hypertrophy or prior MI **(ECG)**

6(O) BUN bag:
Elevated **BUN** & Creatinine levels may suggest renal parenchymal disease causing secondary HTN **(Labs)**

67(O) Credit cards:
Elevated BUN & **Creatinine** levels may suggest renal parenchymal disease causing secondary HTN **(Labs)**

68(O) Bullhorn blasting sound waves at bunches of balloons:
Renal ultrasound should be ordered if renal parenchymal disease (e.g., polycystic kidney disease) is suspected **(Imaging)**

69(O) Salt crystals on ceiling:
In renal artery stenosis (a form of secondary hyperaldosteronism), both **aldosterone levels** and renin activity **will be elevated** → normal aldosterone:renin ratio **(Labs)**

70(O) Yellow umbrella in the air:
In renal artery stenosis (a form of secondary hyperaldosteronism), both aldosterone and **renin activity will be elevated** → normal aldosterone:renin ratio **(Labs)**

71(O) Black cat with contrast collar:
In patients with suspected renal artery stenosis, confirm the diagnosis with imaging of the renal arteries (e.g., **CT angiogram**, Doppler ultrasound, MR angiogram) **(Imaging)**

72(O) Banana peel on floor:
In patients with refractory HTN, hypernatremia, **hypokalemia**, & metabolic alkalosis suggest hyperaldosteronism **(Labs)**

73(O) Basic cleaner raised high:
In patients with refractory HTN, hypernatremia, hypokalemia, & **metabolic alkalosis** suggest hyperaldosteronism **(Labs)**

74(O) Crystal chandelier up near ceiling:
In primary aldosteronism, **aldosterone is elevated**while plasma renin activity is low → aldosterone:renin ratio is elevated (typically >20) **(Labs)**

75(O) Dropped yellow umbrella:
In primary aldosteronism, aldosterone is elevated while **plasma renin activity is low** → aldosterone:renin ratio is elevated (typically >20) **(Labs)**

76(O) “20”-shaped engagement ring:
In primary aldosteronism, aldosterone is elevated while plasma renin activity is low → **aldosterone:renin ratio is elevated (typically >20) (Labs)**

77(O) Dessert inDEX:
If Cushing’s syndrome is suspected, confirm diagnosis with **dexamethasone suppression test**, 24-hour urinary free cortisol, and/or late-night salivary cortisol **(Labs)**

78(O) Mint leaf on yellow ice cream:
If pheochromocytoma is suspected, make the diagnosis by measuring **24-hour urine fractionated metanephrines & catecholamines (Labs)**

79(O) TS Haberdashers bow ties:
Serum **TSH level** will be abnormal in thyroid disease **(Labs)**

80(O) Calci-yum ice cream held up high:
Serum calcium level will be elevated in hyperparathyroidism **(Labs)**

81(O) Skull & X-bones flag:
If coarctation of the aorta is suspected, order **CXR** to look for rib notching **(Imaging)**

82(O) Ladder with bent rungs:
If coarctation of the aorta is suspected, order CXR to look for **rib notching (Imaging)**

83(O) Heart-shaped megaphone:
Confirm diagnosis of aortic coarctation with **transthoracic echo (Imaging)**